

## 1 Summary

Model (Run-name): model1 (test)

Iteration: 6

**TAC: 151684.0776945542**

Took: 1.9185597896575928s

Total Time: 3.972318649291992s

Epsilons:

- $\varepsilon_{LMTD}$ : 1e-06
- $\varepsilon_{Balancing}$ : 1e-06
- $\varepsilon_{Area}$ : 1e-06
- $\varepsilon_{Beta}$ : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: [http://www.gamsworld.org/minlp/minlplib2/html/heatexch\\_gen1.html](http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html)

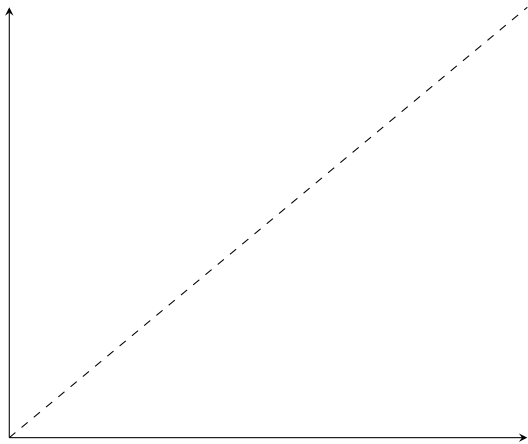
## 2 Results

| Variable          | Value              | Variable          | Value                | Variable            | Value              |
|-------------------|--------------------|-------------------|----------------------|---------------------|--------------------|
| $z_{1,1,1}$       | 1.0                | $dt_{cu,1}$       | 114.74935767560734   | $t_{2,2}^{(H)}$     | 590.0              |
| $z_{1,1,2}$       | 1.0                | $dt_{cu,2}$       | 125.07263782598807   | $t_{2,3}^{(H)}$     | 445.07263782598807 |
| $z_{1,2,2}$       | 1.0                | $dt_{hu,1}$       | 63.263088885055716   | $t_{1,1}^{(C)}$     | 616.7369111149443  |
| $z_{2,1,2}$       | 1.0                | $f_{1,1,1}^H$     | 2.4924738091131116   | $t_{1,2}^{(C)}$     | 570.2107333448329  |
| $z_{2,2,2}$       | 1.0                | $f_{1,1,2}^H$     | 5.199785594067834    | $t_{1,3}^{(C)}$     | 410.0              |
| $z_{cu,1}$        | 1.0                | $f_{1,2,2}^H$     | 4.800214405932166    | $t_{2,1}^{(C)}$     | 500.0              |
| $z_{cu,2}$        | 1.0                | $f_{2,1,2}^H$     | 11.570012031959754   | $t_{2,2}^{(C)}$     | 500.0              |
| $z_{hu,1}$        | 1.0                | $f_{2,2,2}^H$     | 8.429987968040246    | $t_{2,3}^{(C)}$     | 350.0              |
| $A_{1,1,1}^\beta$ | 69.89902342903967  | $f_{1,1,1}^C$     | 14.779954726310777   | $t_{1,1,1}^H$       | 370.0              |
| $A_{1,1,2}^\beta$ | 4.620559768234002  | $f_{1,1,2}^C$     | 2.3102798841170014   | $t_{1,1,2}^H$       | 557.1548250268777  |
| $A_{1,2,2}^\beta$ | 6.000977230027827  | $f_{1,2,2}^C$     | 6.000977230027825    | $t_{1,2,2}^H$       | 395.0              |
| $A_{2,1,2}^\beta$ | 87.16107999267206  | $f_{2,1,2}^C$     | 12.689720115882999   | $t_{2,1,2}^H$       | 404.8354409523366  |
| $A_{2,2,2}^\beta$ | 12.209110369738774 | $f_{2,2,2}^C$     | 6.999022769972175    | $t_{2,2,2}^H$       | 466.1689115177884  |
| $A_{cu,1}^\beta$  | 13.56934656224007  | $RecLMTD_{1,1,1}$ | 0.051651332234625924 | $t_{1,1,1}^C$       | 617.507387405553   |
| $A_{cu,2}^\beta$  | 30.189424467922265 | $RecLMTD_{1,1,2}$ | 0.05905224156032517  | $t_{1,1,2}^C$       | 549.7862368345166  |
| $A_{hu,1}^\beta$  | 13.244985069124171 | $RecLMTD_{1,2,2}$ | 0.048494322187147436 | $t_{1,2,2}^C$       | 500.0              |
| $A_{1,1,1}$       | 69.89902342903967  | $RecLMTD_{2,1,2}$ | 0.034770557092936916 | $t_{2,1,2}^C$       | 542.3940120774585  |
| $A_{1,1,2}$       | 4.620559768234002  | $RecLMTD_{2,2,2}$ | 0.010806066271654577 | $t_{2,2,2}^C$       | 440.01120358475225 |
| $A_{1,2,2}$       | 6.000977230027827  | $RecLMTD_{cu,1}$  | 0.011027825740834664 | $t_{1,1,1}^{H,in}$  | 1620.1079759235224 |
| $A_{2,1,2}$       | 87.16107999267206  | $RecLMTD_{cu,2}$  | 0.010536727983684102 | $t_{1,1,2}^{H,in}$  | 3005.8760586009494 |
| $A_{2,2,2}$       | 12.209110369738774 | $RecLMTD_{hu,1}$  | 0.02242905867850707  | $t_{1,2,2}^{H,in}$  | 2796.231274847379  |
| $A_{cu,1}$        | 13.56934656224007  | $q_{1,1,1}$       | 697.8926665516713    | $t_{2,1,2}^{H,in}$  | 6826.307098856255  |
| $A_{cu,2}$        | 30.189424467922265 | $q_{1,1,2}$       | 554.4671721880802    | $t_{2,2,2}^{H,in}$  | 4973.692901143744  |
| $A_{hu,1}$        | 13.244985069124171 | $q_{1,2,2}$       | 900.1465845041739    | $t_{1,1,1}^{H,out}$ | 922.2153093718513  |
| $dt_{1,1,1}$      | 33.263088885055716 | $q_{2,1,2}$       | 1848.6938279844128   | $t_{1,1,2}^{H,out}$ | 2451.408886412869  |
| $dt_{1,1,2}$      | 10.0               | $q_{2,2,2}$       | 1049.853415495826    | $t_{1,2,2}^{H,out}$ | 1896.084690343205  |
| $dt_{1,1,3}$      | 24.749357675607342 | $q_{cu,1}$        | 647.4935767560734    | $t_{2,1,2}^{H,out}$ | 4977.613270871842  |
| $dt_{1,2,2}$      | 80.21073334483287  | $q_{cu,2}$        | 1501.4527565197614   | $t_{2,2,2}^{H,out}$ | 3923.839485647918  |
| $dt_{1,2,3}$      | 84.74935767560734  | $q_{hu,1}$        | 498.94633327583574   | $t_{1,1,1}^{C,in}$  | 8428.83856246485   |
| $dt_{2,1,2}$      | 19.789266655167093 | $t_{1,1}^{(H)}$   | 650.0                | $t_{1,1,2}^{C,in}$  | 947.2147524879706  |
| $dt_{2,1,3}$      | 35.07263782598806  | $t_{1,2}^{(H)}$   | 580.2107333448329    | $t_{1,2,2}^{C,in}$  | 2100.342030509739  |
| $dt_{2,2,2}$      | 90.0               | $t_{1,3}^{(H)}$   | 434.74935767560737   | $t_{2,1,2}^{C,in}$  | 5202.785247512029  |
| $dt_{2,2,3}$      | 95.07263782598807  | $t_{2,1}^{(H)}$   | 590.0                | $t_{2,2,2}^{C,in}$  | 2449.657969490261  |

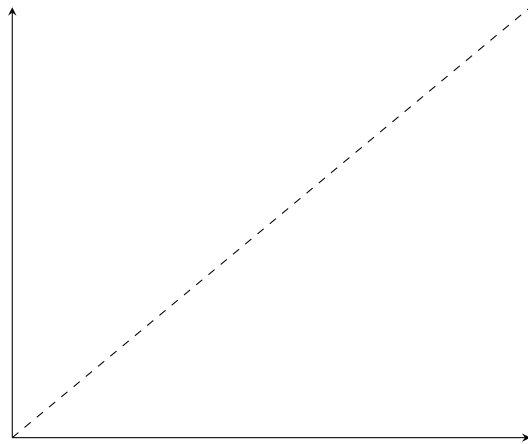
| Variable                   | Value              |
|----------------------------|--------------------|
| $b_{1,1,1}^{C,\text{out}}$ | 9126.731229016523  |
| $b_{1,1,2}^{C,\text{out}}$ | 1501.6819246760506 |
| $b_{1,2,2}^{C,\text{out}}$ | 3000.4886150139127 |
| $b_{2,1,2}^{C,\text{out}}$ | 7051.4790754964415 |
| $b_{2,2,2}^{C,\text{out}}$ | 3499.5113849860877 |

### 3 *RecLMTD* Tangent Points

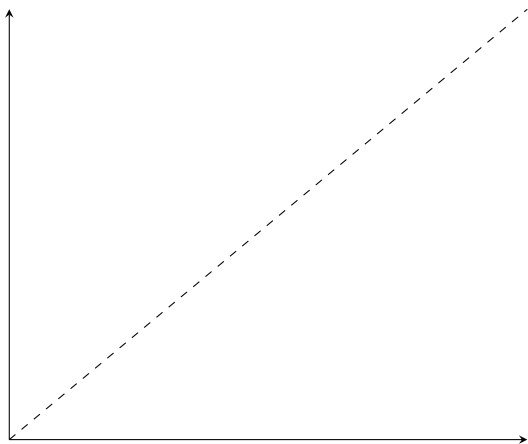
- $(1, 1, 1)$ 
  - Value: 0.051651
  - Error: 0.000012625774758
  - Relative error: 0.000244382645938
- $(1, 1, 2)$ 
  - Value: 0.059052
  - Error: 0.002388701368456
  - Relative error: 0.038878006335688
- $(1, 2, 2)$ 
  - Value: 0.048494
  - Error: 0.036367116887917
  - Relative error: -2.998804422831281
- $(2, 1, 2)$ 
  - Value: 0.034771
  - Error: 0.002674164255330
  - Relative error: 0.071416321421064
- $(2, 2, 2)$ 
  - Value: 0.010806
  - Error: 0.000003208839544
  - Relative error: 0.000296859827413
- $cu, 1$ 
  - Value: 0.011028
  - Error: 0.000017137470936
  - Relative error: 0.001551609598610
- $cu, 2$ 
  - Value: 0.010537
  - Error: 0.000002070446854
  - Relative error: 0.000196459479463
- $hu, 1$ 
  - Value: 0.022429
  - Error: 0.000001349573985
  - Relative error: 0.000060167161024



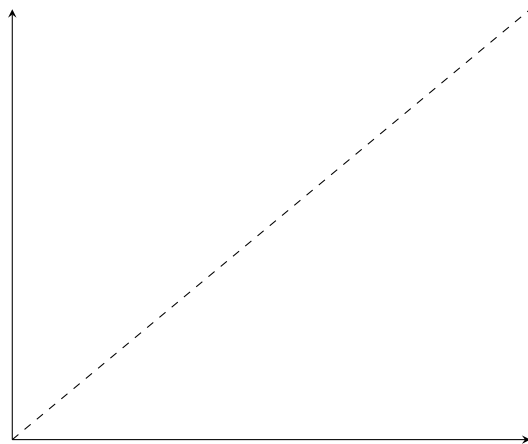
(a) *RecLMTD* - (1, 1, 1)



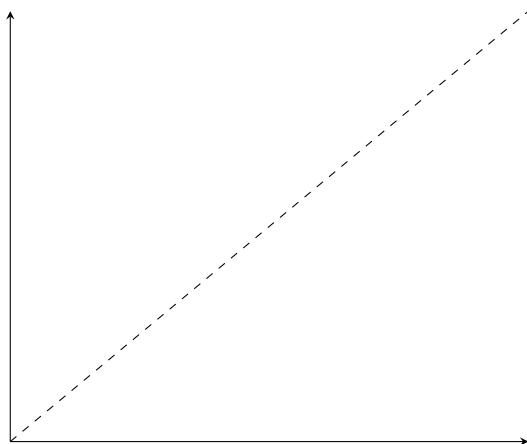
(b) *RecLMTD* - (1, 1, 2)



(c) *RecLMTD* - (1, 2, 2)



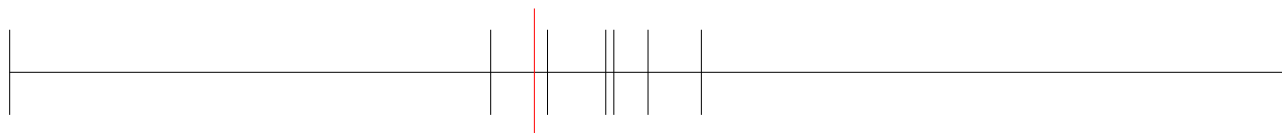
(d) *RecLMTD* - (2, 1, 2)



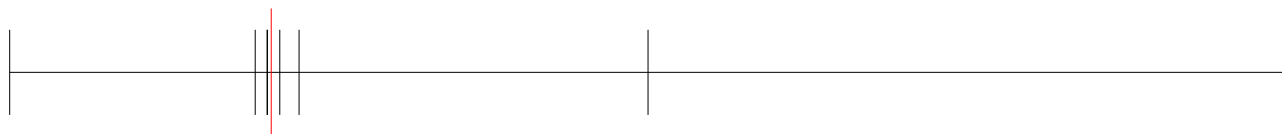
(e) *RecLMTD* - (2, 2, 2)



(a) *RecLMTD* -  $cu, 1$



(b) *RecLMTD* - *cu*, 2



(a) *RecLMTD* - *hu*, 1

## 4 Balancing Breakpoints

### 4.1 $b^{H,\text{in}}$

- (1, 1, 1)
  - Value: 1620.1079759235224
  - Absolute error: 0.0
  - Relative error: 0.0
- (1, 1, 2)
  - Value: 3005.8760586009494
  - Absolute error: 11.095354169046459
  - Relative error: -0.003677646437776284
- (1, 2, 2)
  - Value: 2796.231274847379
  - Absolute error: 11.095354169046004
  - Relative error: 0.00398377475464238
- (2, 1, 2)
  - Value: 6826.307098856255
  - Absolute error: 0.0
  - Relative error: 0.0
- (2, 2, 2)
  - Value: 4973.692901143744
  - Absolute error: 9.094947017729282e-13
  - Relative error: -1.8286104909367883e-16

### 4.2 $b^{H,\text{out}}$

- (1, 1, 1)
  - Value: 922.2153093718513
  - Absolute error: 0.0
  - Relative error: 0.0
- (1, 1, 2)
  - Value: 2451.408886412869
  - Absolute error: 445.6767464272748
  - Relative error: -0.15383623506853603
- (1, 2, 2)
  - Value: 1896.084690343205
  - Absolute error: 4.547473508864641e-13
  - Relative error: -2.398349362781635e-16
- (2, 1, 2)

- Value: 4977.613270871842
- Absolute error: 293.66234808957506
- Relative error: 0.06269543659418608
- (2, 2, 2)
  - Value: 3923.839485647918
  - Absolute error: 5.9588295214562095
  - Relative error: -0.0015163194249574062

### 4.3 $b^{C,in}$

- (1, 1, 1)
  - Value: 8428.83856246485
  - Absolute error: 1.1497391717548453
  - Relative error: 0.00013642401800326414
- (1, 1, 2)
  - Value: 947.2147524879706
  - Absolute error: 0.0
  - Relative error: 0.0
- (1, 2, 2)
  - Value: 2100.342030509739
  - Absolute error: 0.0
  - Relative error: 0.0
- (2, 1, 2)
  - Value: 5202.785247512029
  - Absolute error: 9.094947017729282e-13
  - Relative error: -1.7480919517249885e-16
- (2, 2, 2)
  - Value: 2449.657969490261
  - Absolute error: 0.0
  - Relative error: 0.0

### 4.4 $b^{C,out}$

- (1, 1, 1)
  - Value: 9126.731229016523
  - Absolute error: 0.0
  - Relative error: 0.0
- (1, 1, 2)
  - Value: 1501.6819246760506
  - Absolute error: 231.5218411528815



- Relative error: 0.18227768621943022
- (1, 2, 2)
  - Value: 3000.4886150139127
  - Absolute error: 4.547473508864641e-13
  - Relative error: 1.5155776582886838e-16
- (2, 1, 2)
  - Value: 7051.4790754964415
  - Absolute error: 168.6508697026311
  - Relative error: 0.0245031351444548
- (2, 2, 2)
  - Value: 3499.5113849860877
  - Absolute error: 419.8629520535442
  - Relative error: 0.13633470222240165



(a)  $t^{(H)} - (1, 1)$



(a)  $t^{(H)} - (1, 2)$



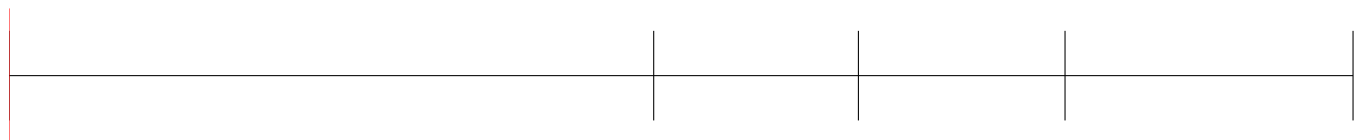
(a)  $t^{(H)} - (1, 2)$



(a)  $t^{(H)} - (2, 2)$



(a)  $t^{(H)} - (2, 2)$



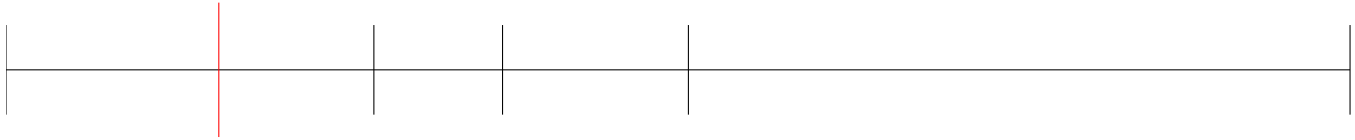
(a)  $t^{(H)} - (1, 1, 1)$



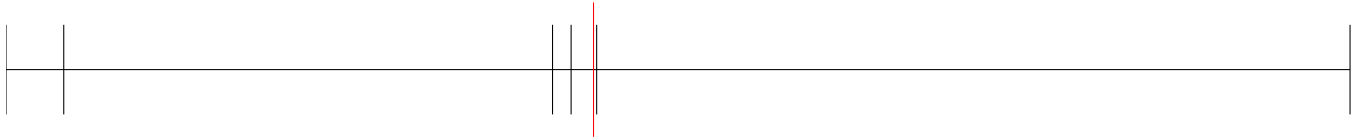
(a)  $t^{(H)} - (1, 1, 2)$



(a)  $t^{(H)} - (1, 2, 2)$



(a)  $t^{(H)} - (2, 1, 2)$



(a)  $t^{(H)} - (2, 2, 2)$



(a)  $t^{(C)} - (1, 2)$



(a)  $t^{(C)} - (1, 3)$



(a)  $t^{(C)} - (2, 3)$



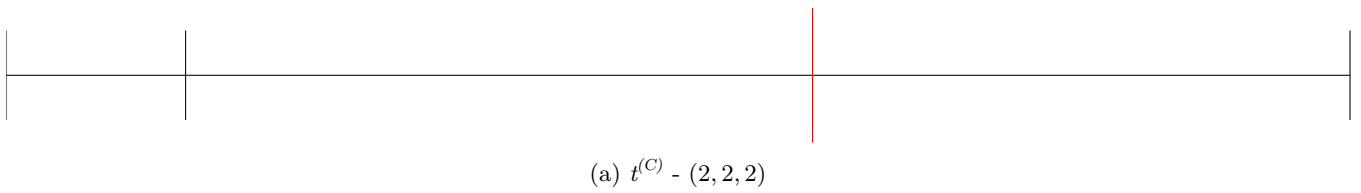
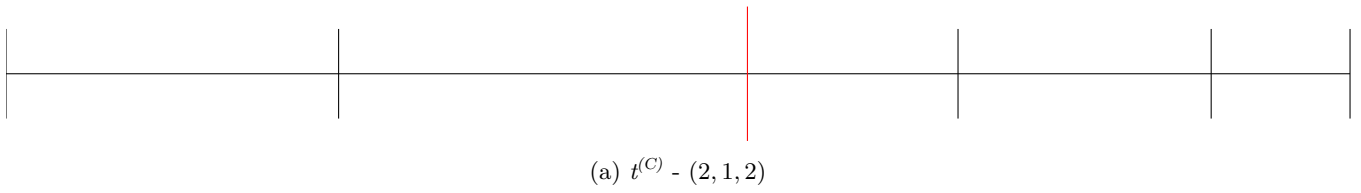
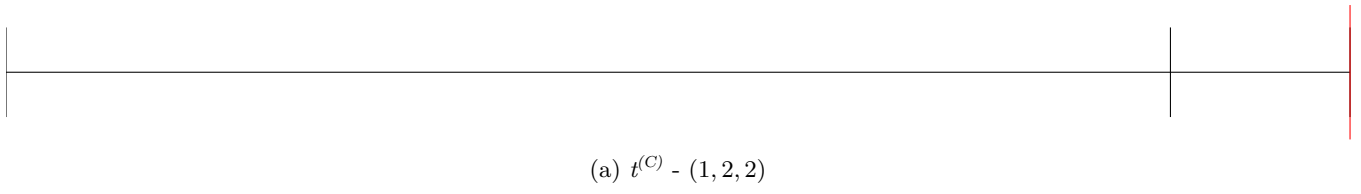
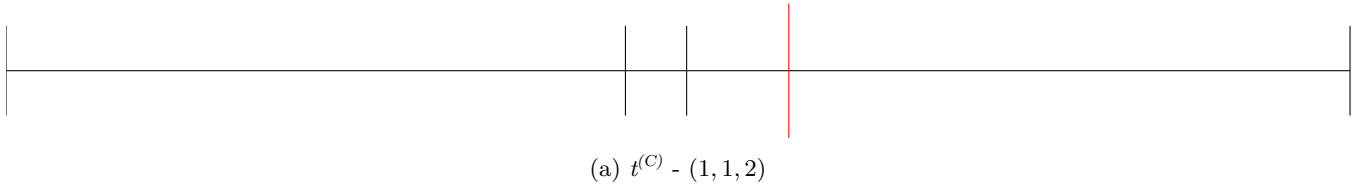
(a)  $t^{(C)} - (1, 3)$



(a)  $t^{(C)} - (2, 3)$



(a)  $t^{(C)} - (1, 1, 1)$



## 5 Area Breakpoints

### 5.1 $A$

- $(1, 1, 1)$ 
  - Value: 69.899023
  - Error: 2.195148539299083
  - Relative error: 0.030448349420854
- $(1, 1, 2)$ 
  - Value: 4.620560
  - Error: 60.864499010407840
  - Relative error: 0.929441007545663
- $(1, 2, 2)$ 
  - Value: 6.000977
  - Error: 81.303019739183668
  - Relative error: 0.931263430789496
- $(2, 1, 2)$ 
  - Value: 87.161080
  - Error: 41.399148593912201
  - Relative error: 0.322021429559222
- $(2, 2, 2)$ 
  - Value: 12.209110
  - Error: 10.480460797002834
  - Relative error: 0.461906517315105

### 5.2 $A_{CU}$

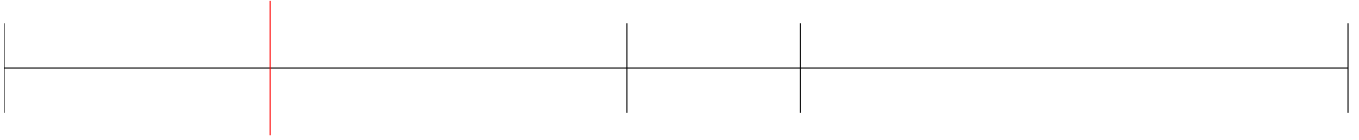
- $cu, 1$ 
  - Value: 13.569347
  - Error: 0.711546103311392
  - Relative error: 0.049825043852321
- $cu, 2$ 
  - Value: 30.189424
  - Error: 1.451374083680538
  - Relative error: 0.045870336720911

### 5.3 $A_{HU}$

- $hu, 1$ 
  - Value: 13.244985
  - Error: 0.184090834639427
  - Relative error: 0.013708376954503



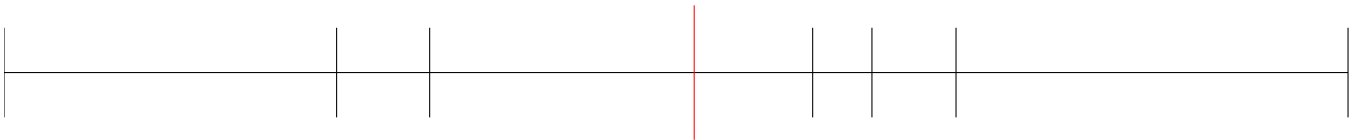
(a)  $A^\beta - (1, 1, 1)$



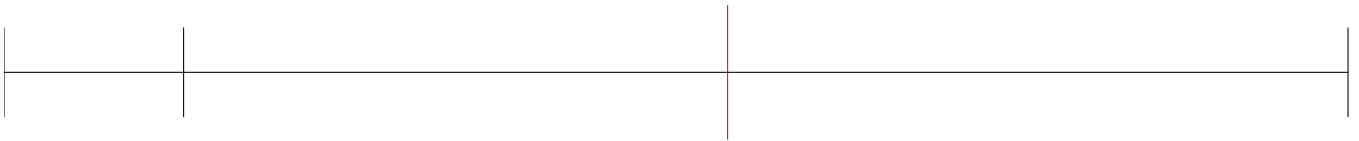
(a)  $A^\beta - (1, 1, 2)$



(a)  $A^\beta - (1, 2, 2)$



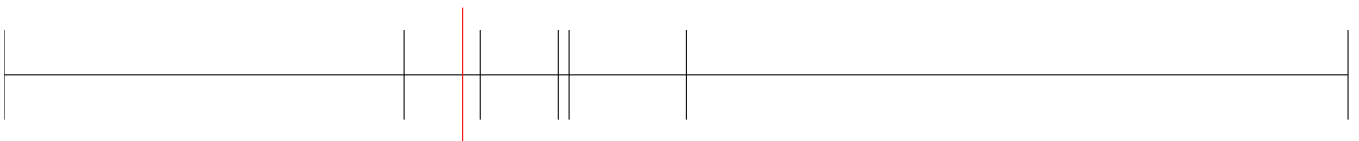
(a)  $A^\beta - (2, 1, 2)$



(a)  $A^\beta - (2, 2, 2)$



(a)  $A^\beta - cu, 1$



(a)  $A^\beta - cu, 2$



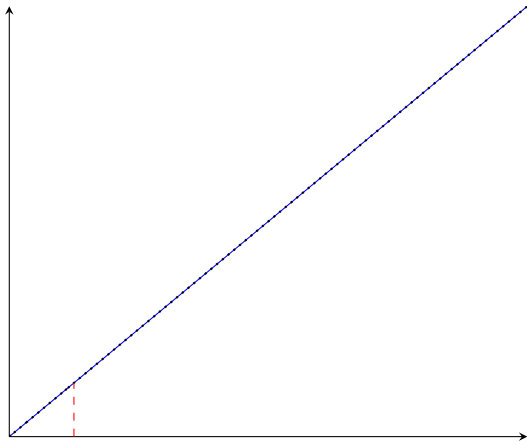
(a)  $A^\beta - hu, 1$

## 6 $A^\beta$ Breakpoints

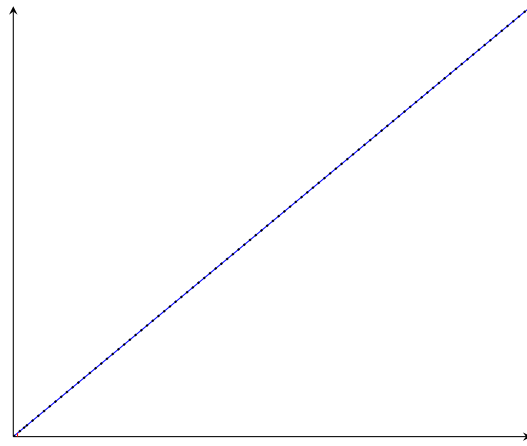
- (1, 1, 1)
  - $q$ : 69.899023
  - $\hat{q}^\beta$ : 69.899023
  - $q^\beta$ : 69.899023
  - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 69.14107665413653, 560.0]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- (1, 1, 2)
  - $q$ : 4.620560
  - $\hat{q}^\beta$ : 4.620560
  - $q^\beta$ : 4.620560
  - Breakpoints: [0, 10.813867750736755, 13.825772197515043, 560.0]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- (1, 2, 2)
  - $q$ : 6.000977
  - $\hat{q}^\beta$ : 6.000977
  - $q^\beta$ : 6.000977
  - Breakpoints: [0, 11.262728503004022, 39.36524453694061, 84.1468297226002, 390.0]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- (2, 1, 2)
  - $q$ : 87.161080
  - $\hat{q}^\beta$ : 87.161080
  - $q^\beta$ : 87.161080
  - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 720.0]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- (2, 2, 2)
  - $q$ : 12.209110
  - $\hat{q}^\beta$ : 12.209110
  - $q^\beta$ : 12.209110
  - Breakpoints: [0, 2.171589371244973, 38.943261778500734, 44.259377040939995, 390.0]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000

- $cu, 1$ 
  - $q$ : 13.569347
  - $\hat{q}^\beta$ : 13.569347
  - $q^\beta$ : 13.569347
  - Breakpoints: [0, 2.9819181007907165, 7.702824927089129, 9.28003023700696, 14.535545040173215, 181.61828057849593]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- $cu, 2$ 
  - $q$ : 30.189424
  - $\hat{q}^\beta$ : 30.189424
  - $q^\beta$ : 30.189424
  - Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 34.23465128608717, 38.88483572219945, 285.4001551947793]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000
- $hu, 1$ 
  - $q$ : 13.244985
  - $\hat{q}^\beta$ : 13.244985
  - $q^\beta$ : 13.244985
  - Breakpoints: [0, 4.943755299006499, 12.595096067947068, 13.192252946941991, 13.30871858348944, 14.09178614751222, 237.3002543523117]
  - Error: 0.0000000000000000
  - Relative error: 0.0000000000000000

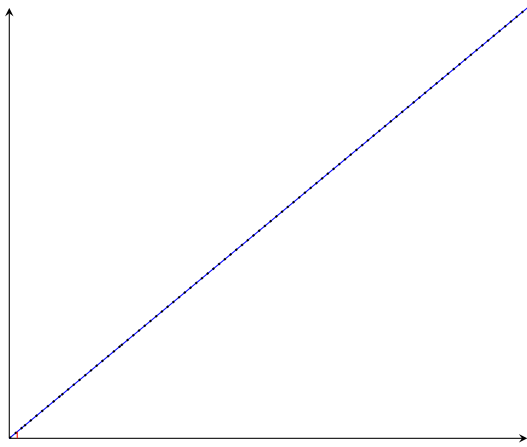




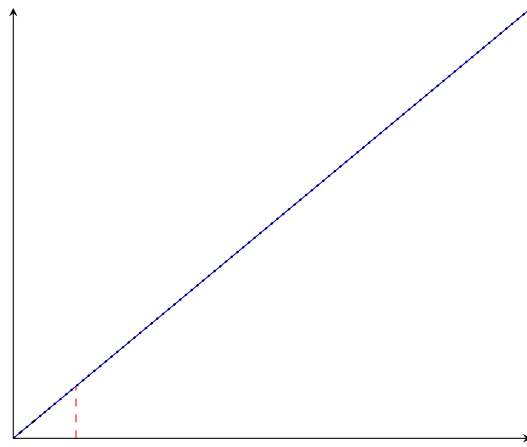
(a)  $q^\beta - (1, 1, 1)$



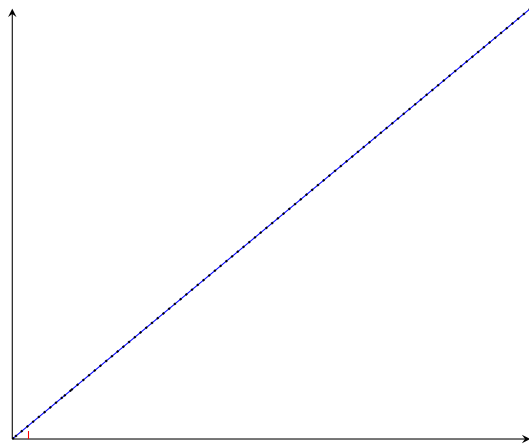
(b)  $q^\beta - (1, 1, 2)$



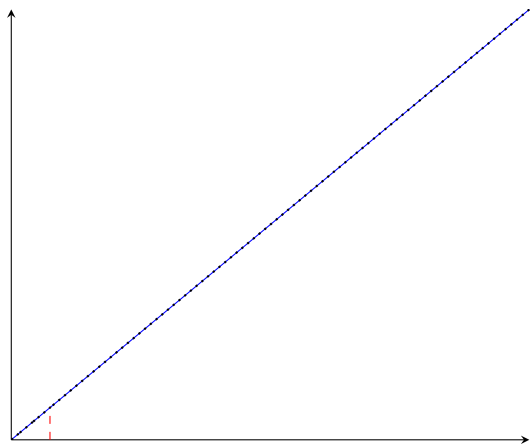
(c)  $q^\beta - (1, 2, 2)$



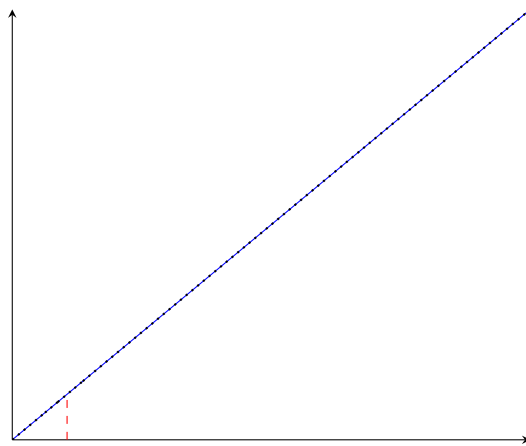
(d)  $q^\beta - (2, 1, 2)$



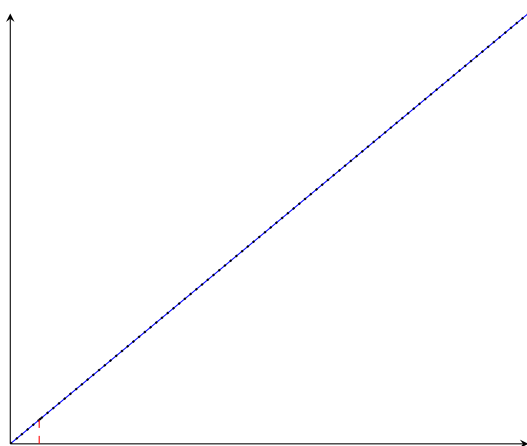
(e)  $q^\beta - (2, 2, 2)$



(a)  $q^\beta - cu, 1$



(b)  $q^\beta - cu, 2$



(a)  $q^\beta - hu, 1$

## 7 Other variables

| Variable          | Value                | Variable          | Value              |
|-------------------|----------------------|-------------------|--------------------|
| $z_{1,2,1}$       | 0.0                  | $C_{1,2,1}$       | 500.0              |
| $z_{2,1,1}$       | 0.0                  | $C_{2,1,1}$       | 412.2735916996707  |
| $z_{2,2,1}$       | 0.0                  | $C_{2,2,1}$       | 350.0              |
| $z_{hu,2}$        | -0.0                 | $H_{1,2,1}^{in}$  | 4879.892024076477  |
| $A_{1,2,1}^\beta$ | 0.0                  | $H_{2,1,1}^{in}$  | 11800.0            |
| $A_{2,1,1}^\beta$ | 0.0                  | $H_{2,2,1}^{in}$  | 0.0                |
| $A_{2,2,1}^\beta$ | 0.0                  | $H_{1,2,1}^{out}$ | 4879.892024076477  |
| $A_{hu,2}^\beta$  | 0.0                  | $H_{2,1,1}^{out}$ | 11800.0            |
| $A_{1,2,1}$       | 0.0                  | $H_{2,2,1}^{out}$ | 0.0                |
| $A_{2,1,1}$       | 0.0                  | $C_{1,2,1}^{in}$  | 6500.0             |
| $A_{2,2,1}$       | 0.0                  | $C_{2,1,1}^{in}$  | 124.32243770764211 |
| $A_{hu,2}$        | 0.0                  | $C_{2,2,1}^{in}$  | 0.0                |
| $dt_{1,2,1}$      | 150.0                | $C_{1,2,1}^{out}$ | 6500.0             |
| $dt_{2,1,1}$      | 14.377400011499574   | $C_{2,1,1}^{out}$ | 124.32243770764211 |
| $dt_{2,2,1}$      | 90.0                 | $C_{2,2,1}^{out}$ | 0.0                |
| $dt_{hu,2}$       | 180.0                |                   |                    |
| $f_{1,2,1}^H$     | 7.507526190886889    |                   |                    |
| $f_{2,1,1}^H$     | 20.0                 |                   |                    |
| $f_{2,2,1}^H$     | 0.0                  |                   |                    |
| $f_{1,2,1}^C$     | 13.0                 |                   |                    |
| $f_{2,1,1}^C$     | 0.2200452736892231   |                   |                    |
| $f_{2,2,1}^C$     | 0.0                  |                   |                    |
| $RecLMTD_{1,2,1}$ | 0.008112159555778712 |                   |                    |
| $RecLMTD_{2,1,1}$ | 0.029166666666666688 |                   |                    |
| $RecLMTD_{2,2,1}$ | 0.010239999999999999 |                   |                    |
| $RecLMTD_{hu,2}$  | 0.005549329826813255 |                   |                    |
| $q_{1,2,1}$       | 0.0                  |                   |                    |
| $q_{2,1,1}$       | 0.0                  |                   |                    |
| $q_{2,2,1}$       | 0.0                  |                   |                    |
| $q_{hu,2}$        | 0.0                  |                   |                    |
| $t_{1,2,1}^H$     | 650.0                |                   |                    |
| $t_{2,1,1}^H$     | 590.0                |                   |                    |
| $t_{2,2,1}^H$     | 590.0                |                   |                    |